$$= \lim_{\Delta x \to 0} \left[\frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} \right] \times \lim_{\Delta x \to 0} (\Delta x)$$

$$= f'(x_0) \times 0 = 0 .$$

This implies, f is continuous at $x = x_0$. ■

Derivatives from first principle

The process of finding the derivative of a function using the conditions stated in the definition of derivatives is known as derivatives from first principle.

EXERCISE 10.1

(1) Find the derivatives of the following functions using first principle.

(i) $f(x) = 6$ (ii) $f(x) = -4x + 7$ (iii) $f(x) = -x^2 + 2$

(2) Find the derivatives from the left and from the right at $x = 1$ (if they exist) of the following functions. Are the functions differentiable at $x = 1$?

(i) $f(x) = |x - 1|$ (ii) $f(x) = \sqrt{1 - x^2}$ (iii) $f(x) = \begin{cases} x, & x \le 1 \\ x^2, & x > 1 \end{cases}$

(3) Determine whether the following function is differentiable at the indicated values.

(i) $f(x) = x\,|x|$ at $x = 0$ (ii) $f(x) = |x^2 - 1|$ at $x = 1$

(iii) $f(x) = |x| + |x - 1|$ at $x = 0, 1$ (iv) $f(x) = \sin|x|$ at $x = 0$

(4) Show that the following functions are not differentiable at the indicated value of x.

(i) $f(x) = \begin{cases} -x + 2, & x \le 2 \\ 2x - 4, & x > 2 \end{cases}$; $x = 2$ (ii) $f(x) = \begin{cases} 3x, & x < 0 \\ -4x, & x \ge 0 \end{cases}$, $x = 0$

(5) The graph of f is shown below. State with reasons that x values (the numbers), at which f is not differentiable.

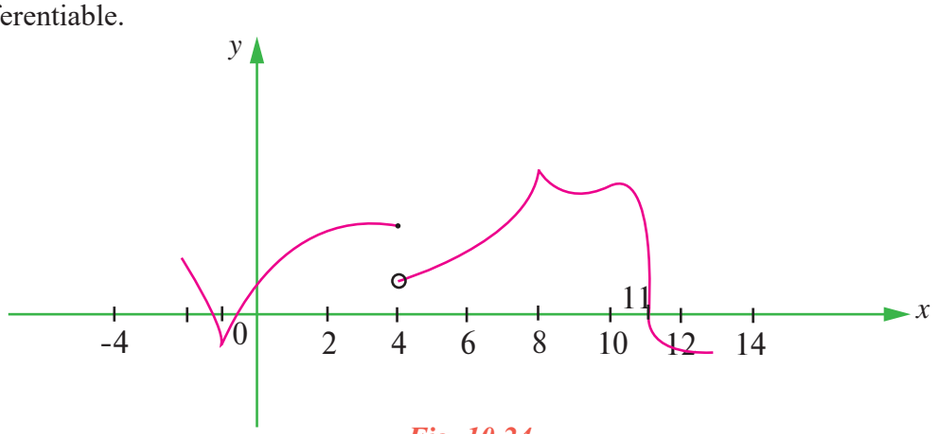


Fig. 10.24

(6) If $f(x) = |x + 100| + x^2$, test whether $f'(-100)$ exists.

(7) Examine the differentiability of functions in $\mathbb{R}$ by drawing the diagrams.

(i) $|\sin x|$ (ii) $|\cos x|$.

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